Feature Description 83/221 04-LZA 701 6014/1 Uen ED

# Inter-Frequency Load Balancing

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# 1 Inter-Frequency Load Balancing Overview

This feature manages the uneven distribution of traffic load between different carrier frequencies.

| | |
|---|---|
| Feature Name | Inter-Frequency Load Balancing |
| Feature Identity | FAJ 121 3009 |
| Value Package Name | Multi-Carrier Load Management |
| Value Package Identity | FAJ 801 0427 |
| Node Type | Baseband Radio Node |
| Access Type | LTE |
| Licensing | License-controlled feature. One license is required for each node. |

## Summary

Inter-frequency load balancing enables the efficient use of network resources on multiple carrier frequencies, and achieves similar user experience independent of the carrier in use. Load balancing is achieved by relocating the UE in connected mode to carriers that are underused compared to the carrier in use.

This feature introduces a method to assess traffic load. The method is based on the presence of E-RABs in the cell and the QoS class to which they belong.

The Inter-Frequency Load Balancing feature is realized as a network feature. Neighboring nodes cooperate and perform the traffic load adjustments without a central node controlling those efforts.

The Inter-Frequency Load Balancing feature can be inter-mode (FDD and TDD) if the Intra-LTE Inter-Mode Handover feature is activated.

The Inter-Frequency Load Balancing feature supports the following functionalities:

- An arbitrary grade of coverage overlap between cells
- An arbitrary number of target cells on the same inter-frequency
- An arbitrary number of frequencies and frequency bands
- Load balancing between cells that are co-located and not co-located
- Load balancing between cells in the same and in separate nodes

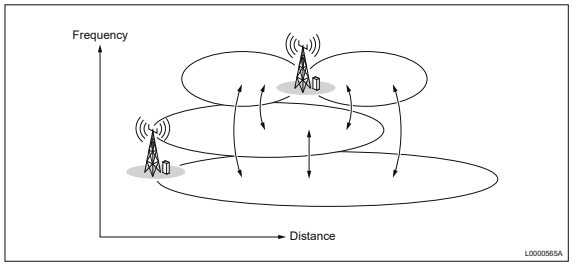



Figure 1 Traffic Moving between Inter-Frequency Cells Covering a Common Geographical Area Due to Load Balancing

To support load balancing between cells in separate nodes, the feature requires an Ericsson node in both ends.

The feature provides the following benefits:

- Radio traffic load is distributed between inter-frequency cells with overlapping coverage.

  For example, lower carrier frequencies can have better coverage and propagation characteristics and attract more traffic than higher frequencies. This can result in the underuse of high frequency carriers. The Inter-Frequency Load Balancing feature distributes radio use evenly over all carriers, which maximizes the availability of radio resources.
- The non-GBR end user experience is enhanced.
- The risk of blocking is reduced, allowing more UEs in an area where multiple carrier frequencies are used.
- The average user throughput increases linearly when adding new carrier frequencies or increasing the RF spectrum by other means.
- The observability of the traffic load in the network increases on a per-cell basis using a subscription ratio.

### Additional Information

For information on the preferred state and parameter settings of this feature, see RAN Parameter Recommendations Lists.
For more information about this feature and related topics, see the following documentation:

- *3GPP TS 36.300, Overall description; Stage 2*
- *3GPP TS 36.331, Radio Resource Control (RRC); Protocol Specification*
- *3GPP TS 36.423, X2 Application Protocol (X2AP)*
- Admission-Triggered Offload
- Automated Cell Capacity Estimation
- Best Neighbor Relations for Intra-LTE Load Management
- Best Neighbor Relations for WCDMA IRAT Offload
- Coverage-Adapted Load Management
- Coverage-Triggered Inter-Frequency Handover
- Coverage-Triggered Inter-Frequency Session Continuity
- Dynamic GBR Admission Control
- Idle Mode Support
- Inter-Frequency Offload
- Inter-RAT Offload to WCDMA
- Limited-Uplink-Aware IFLB
- Overlaid Cell Detection
- Preferential Traffic Management Guidelines
- Radio Bearer Service
- Service Specific Load Management
- Service Triggered Mobility
- Shared LTE RAN
- Subscriber Triggered Mobility
- UE Throughput-Aware IFLB

# 2 Dependencies of Inter-Frequency Load Balancing

### Features

Table 1 Feature Dependencies

| Feature | Relationship | Description |
|---|---|---|
| Coverage-Triggered Inter-Frequency Handover (FAJ 121 0877) | Prerequisite | Coverage-Triggered Inter-Frequency Handover must be active prior to introducing the Inter-Frequency Load Balancing feature. Handover is triggered based on inter-frequency load balancing. |
| Accelerated Inter-Frequency Load Balancing (FAJ 121 5036) | Related | The Accelerated Inter-Frequency Load feature makes this feature more configurable. At least one of the following features must be enabled for the Accelerated Inter-Frequency Load feature to operate:<br>– Inter-Frequency Load Balancing<br>– Inter-Frequency Offload<br>– Inter-RAT Offload to WCDMA<br>When uplink-heavy UE detection and exclusion are active in a cell, uplink-heavy UEs are excluded from load balancing mobility. This gives priority to such UEs for the mobility actions of the Uplink-Traffic-Triggered Mobility feature. |
| Admission-Triggered Offload (FAJ 121 3100) | Related | Provides control of the load created by GBR bearers in LTE cells. It offloads the User Equipments (UEs) to other LTE or WCDMA cells when the source cell's GBR usage exceeds the threshold. |
| Automated Cell Capacity Estimation (FAJ 121 3031) | Related | Provides the function to determine and configure cell downlink capacity automatically. |
| Automated Neighbor Relations (FAJ 121 0497) | Related | Activating the ANR Detection of Problematic Cells or ANR Whitelist PLMN IDs functionality prevents load balancing on cell relations which are barred for mobility. |
| Basic Intelligent Connectivity (FAJ 121 4843) | Related | The Inter-Frequency Load Balancing feature can be applied for EN-DC capable UEs with SN terminated bearers and EN-DC capable UEs without SN terminated bearers using the same criteria as for LTE UEs. The load balancing can be configured with the `EUtranCellFDD.lbActionForEndcUe` attribute or the`EUtranCellTDD.lbActionForEndcUe` attribute. |
| Basic Load Management (FAJ 121 3092) | Related | The objective for the load management functions in the eNodeB is to balance or offload load when possible. During the execution of load management functions the MP processor load can increase. To avoid affecting general traffic handling, the load management functions are temporarily paused if the MP load control state is higher than `NOT HIGH`. |
| Best Neighbor Relations for Intra-LTE Load Management (FAJ 121 3028) | Related | Provides the functions to find and evaluate suitable E-UTRAN relations for Inter-frequency load balancing or offload automatically. |
| Best Neighbor Relations for WCDMA IRAT Offload (FAJ 121 3065) | Related | Provides the functions to find and evaluate suitable WCDMA IRAT offload neighbor relations for WCDMA IRAT offload automatically. |
| Carrier Aggregation-Aware IFLB (FAJ 121 3075) | Related | Improves load balancing for CA-capable UEs by distributing them efficiently to cells where the carrier aggregation capability can be utilized in a better way.<br>When uplink-heavy UE detection and exclusion are active in a cell, the UE might be moved to a neighbor cell before it is identified as uplink-heavy. If it is detected as uplink-heavy in the serving cell it was moved to, it might be immediately moved back to the source cell. After the last handover, the UE is identified as uplink-heavy and is not chosen for further load-balancing actions as long as it is marked as uplink-heavy. |
| Carrier Aggregation Downlink Traffic Steering (FAJ 121 5237) | Related | When calculating the subscription ratio, the Inter-Frequency Load Balancing feature only considers the capacity of the PCell and the presence of E-RABs, while the capacity and use of SCells are not considered. When the Inter-Frequency Load Balancing and Carrier Aggregation Downlink Traffic Steering features are used together, the subscription ratio can be affected. This can lead to suboptimal load balancing. |
| Coverage-Adapted Load Management (FAJ 121 3077) | Related | This feature optimizes the load to be transferred to neighboring cells and also the number of required UE measurements for load balancing purposes. It also allows large numbers of load balancing relations to be handled with better efficiency. This facilitates load balancing towards small cells. |

| Feature | Relationship | Description |
|---|---|---|
| Coverage-Triggered Inter-Frequency Session Continuity (FAJ 121 0797) | Related | Feature providing basic connected mode mobility between frequencies in the LTE network. When the feature is active, the UE can be directed to transfer between frequencies in the LTE network while maintaining the data session. |
| Device-Based Frequency Filtering (FAJ 121 5154) | Related | The Device-Based Frequency Filtering feature is used to configure a list of unsupported frequencies for specific UEs. These frequencies are excluded from UE configuration messages sent by the eNodeB.<br>Unsupported frequencies are configured using the `UnsupportedFreqProfile.unsupportedFreqList` attribute. |
| Dynamic GBR Admission Control (FAJ 121 1748) | Related | Dynamic GBR admission control is performed by monitoring the use of certain system resources, referred to as monitored system resources (MSR), and allowing the setting up of new GBR bearers only if the MSR use by GBR bearers does not exceed a configured threshold. |
| Idle Mode Support (Part of LTE Basic) | Related | This feature is part of LTE Basic. In idle mode, the UE has no active connection to the RAN. The Idle Mode Support feature enables the UE in idle mode to access the network and be reached from the network within acceptable delay. |
| Inter-Frequency Offload (FAJ 121 3061) | Related | The purpose of this feature is to off-load traffic load above an off-load threshold from an E-UTRAN cell to an Ericsson or non-Ericsson E-UTRAN FDD or TDD cell. This feature uses the method introduced by Inter-frequency Load Balancing for the assessment of traffic load, based on the presence of E-RAB, in the cell and the QoS class to which they belong.<br>This is an alternative feature to IFLB. The main difference to IFLB is that there is no load information exchange between the cells in Inter-Frequency Offload. |
| Inter-RAT Offload to WCDMA (FAJ 121 3048) | Related | This feature is realized as a network feature. Neighboring LTE and WCDMA cells cooperate and perform the traffic load adjustments, without a central node controlling those efforts. The feature uses the introduced method from Inter-frequency Load Balancing for the assessment of traffic load. Assessment is based on the presence of Evolved Radio Access Bearers (E-RAB) in the cell and the QoS class to which they belong. |
| Intra-LTE Inter-Mode Handover (FAJ 121 3042) | Related | This feature provides the possibility to initiate a handover from an LTE TDD/FDD cell to an LTE FDD/TDD cell. This is beneficial for operators that have an LTE network where TDD and FDD cells coexist in the LTE RAN. |
| Limited-Uplink-Aware IFLB (FAJ 121 4406) | Related | The Limited-Uplink-Aware IFLB feature enhances Inter-Frequency Load Balancing, so that the calculation of cell subscription capacity considers the uplink capacity of the cell. This helps to prevent the limited-uplink cell from uplink congestion during load balancing. |
| MBMS Multi-Carrier Support (FAJ 121 3062) | Related | This feature supports the UE to move to or stay on a carrier that broadcasts an MBMS Service in which the UE is interested. UE interested in eMBMS on serving frequency are not considered for load balancing. |
| Multiple Frequency Band Indicators (FAJ 121 3054) | Related | The feature provides a mechanism for managing the uneven traffic load distribution between different carrier frequencies. The carrier frequency in the other network might be in the same LTE mode or a different mode.<br>If the Multiple Frequency Band Indicators feature is supported, and the UE does not support the primary frequency band, a check for additional frequency bands towards UE capabilities is performed. The aim is to find a possible frequency to configure measurement. If no frequency in additional band is suitable, the UE and the frequency are no considered for Load Balancing. |
| Overlaid Cell Detection (FAJ 121 3078) | Related | The Overlaid Cell Detection feature optimizes detection of small cells as load balancing relations for load balancing or offload by use of automatic and dynamic identification. This facilitates load management towards small cells and potentially enhances the load balancing performance of the Inter-Frequency Load Balancing. |
| Radio Bearer Service (Part of LTE Basic) | Related | This feature is part of LTE Basic. The feature provides the service of establishment, maintenance and release of SRBs and DRBs. |
| Service Specific Load Management (FAJ 121 3047) | Related | Service Specific Load Management enhances control of the UE selection process for Inter-Frequency Load Balancing, Inter-RAT Offload to WCDMA and Inter-Frequency Offload. |
| Service Triggered Mobility (FAJ 121 1747) | Related | This feature enables coverage-triggered mobility based on the QoS defined for the UE bearers. The feature applies dynamic levels of coverage thresholds based on the QCI profiles of the bearers. |
| Shared LTE RAN (FAJ 121 0860) | Related | A core network operator provides its mobile phone services through a PLMN. Conventionally, a PLMN consists of an RAN and a core network, through which only one operator provides services to its subscribers. Subscribers of other operators can receive services as national or international roamers. With the Shared LTE RAN feature, up to six operators that is, with up to six PLMNs, can share an LTE RAN. The Inter-Frequency Load Balancing feature is not supported for eNodeBs in a |

| Feature | Relationship | Description |
|---|---|---|
| | | MORAN on single Baseband configuration. The `EUtranCellRelation.loadBalancing` attribute must be set to `NOT_ALLOWED` on the local and external eNodeB for all cell relations. |
| Subscriber Triggered Mobility (FAJ 121 1788) | Related | If the RFPM function of Subscriber Triggered Mobility is activated and the UE has a SPID value associated with it, the inter-frequency load balancing is inhibited for that UE. It is inhibited only when the `loadBalancingAllowed` attribute related to the concerned `RATFreqPrio` MO for the concerned target carrier frequency is set to `false`. |
| UE Throughput-Aware IFLB (FAJ 121 4219) | Related | UE Throughput-Aware IFLB is an enhancement to the Inter-Frequency Load Balancing.<br>Throughput-aware UE selection estimates and compares possible UE throughput in the cell before using a UE for load balancing. A decision on whether to move the UE or not is based on the throughput difference.<br>When uplink-heavy UE detection and exclusion are active in a cell, the UE might be moved to a neighbor cell before it is identified as uplink-heavy. If it is detected as uplink-heavy in the serving cell it was moved to, it might be immediately moved back to the source cell. After the last handover, the UE is identified as uplink-heavy and is not chosen for further load-balancing actions as long as it is marked as uplink-heavy. |
| PLMN-Specific Access Barring (FAJ 121 4916) | Conflicting | When the Inter-Frequency Load Balancing feature is used together with automatic access class barring, the cell stops broadcasting its load status to other cells. It is removed from the list of candidates for inter-frequency load balancing. This is to avoid handover to an already overloaded cell. The load measurement is ongoing to enable the outgoing handover to other cells, and to decrease the load faster. If load balancing during access class barring is disabled, all inter-frequency load balancing processes are disabled in the cell. |
| Virtual Sectors (FAJ 121 4730) | Conflicting | There are differing compatibility issues between Inter-Frequency Load Balancing and Virtual Sectors in various cases, depending on which feature is activated earlier. Inter-Frequency Load Balancing cannot work between virtual sectors. However Inter-Frequency Load Balancing is still possible between a virtual sector and a normal sector. |
| Progressive Access Barring (FAJ 121 4746) | Related | When the Inter-Frequency Load Balancing feature is used together with automatic access class barring, the cell stops broadcasting its load status to other cells. It is removed from the list of candidates for inter-frequency load balancing. This is to avoid handover to an already overloaded cell. The load measurement is ongoing to enable the outgoing handover to other cells, and to decrease the load faster. If load balancing during access class barring is disabled, all inter-frequency load balancing processes are disabled in the cell. |
| Load-Based Access Barring (FAJ 121 3093) | Related | When the Inter-Frequency Load Balancing feature is used together with automatic access class barring, the cell stops broadcasting its load status to other cells. It is removed from the list of candidates for inter-frequency load balancing. This is to avoid handover to an already overloaded cell. The load measurement is ongoing to enable the outgoing handover to other cells, and to decrease the load faster. If load balancing during access class barring is disabled, all inter-frequency load balancing processes are disabled in the cell. |
| Inter-Vendor Load Balancing (FAJ 121 4946) | Related | The Inter-Vendor Load Balancing feature uses the load balancing algorithms of the Inter-Frequency Load Balancing feature. The Inter-Vendor Load Balancing feature cannot be activated without activating the Inter-Frequency Load Balancing feature. |
| Mission-Critical Push-to-Talk (FAJ 121 4929) | Related | When the Mission-Critical Push-to-Talk feature is active, this feature treats mission-critical bearers like it treats VoLTE bearers.<br>The recommended value for `QciProfilePredefined.qciSubscriptionQuanta` and `QciProfileOperatorDefined.qciSubscriptionQuanta` is 1 for GBR and non-GBR E-RABs with QCIs 65, 66, 69, and 70. |
| Uplink-Traffic-Triggered Mobility (FAJ 121 5044) | Related | When the following features and the Uplink-Traffic-Triggered Mobility feature are enabled at the same time, after an uplink-heavy handover the target eNodeB does not consider the UE for inter-frequency load balancing or offload decision until an internal timer runs out:<br>– Inter-Frequency Load Balancing<br>– Carrier Aggregation-Aware IFLB<br>– Limited-Uplink-Aware IFLB |

| Feature | Relationship | Description |
|---|---|---|
| | | – UE Throughput-Aware IFLB<br>– Inter-Frequency Offload<br>If uplink-heavy detection for load balancing is active for any of these features, uplink-heavy UEs are excluded from load-balancing mobility in the cell. |
| Evolved Load-Based Distribution at Release (FAJ 121 5128) | Related | The load measuring functions of Inter-Frequency Load Balancing are used to find high loaded frequencies in Evolved Load-Based Distribution at Release. |
| Capability-Aware Idle Mode Control (FAJ 121 5073) | Related | The Inter-Frequency Load Balancing feature provides cell load information during the load balancing cycle. This way, frequencies can be prioritized based on cell load using the Capability-Aware Idle Mode Control feature.<br>If the subscription ratio provided in the Inter-Frequency Load Balancing feature is greater than or equal to the value of the `EUtranCellFDD.subscriptionRatioHighThresh` or the `EUtranCellTDD.subscriptionRatioHighThresh` attribute, the cell load is considered high. |
| TM9 Capability-Aware Idle Mode Control (FAJ 121 5153) | Related | The Inter-Frequency Load Balancing feature provides cell load information during the load balancing cycle. This way, frequencies can be prioritized based on cell load using the TM9 Capability-Aware Idle Mode Control feature.<br>If the subscription ratio provided in the Inter-Frequency Load Balancing feature is greater than or equal to the value of the `EUtranCellFDD.subscriptionRatioHighThresh` or the `EUtranCellTDD.subscriptionRatioHighThresh` attribute, the cell load is considered high. |
| Radio Resource Partitioning (FAJ 121 4571) | Conflicting | The Inter-Frequency Load Balancing feature is not partition-aware. This means that the traffic load is distributed between different carrier frequencies without considering the traffic load of the individual partitions. Handovers that are initiated by the Inter-Frequency Load Balancing feature to an overused partition might be rejected if dynamic GBR admission control is enabled on a partition level. Handover to an overused partition in which resources are limited by the `ResourcePartitionGroupMember.resourcePartitionMaxShare` attribute can degrade UE performance. |

### Hardware

No special hardware requirement is expected for this feature.

### Limitations

No limitations for this feature.

### Network Requirements

The cell relations where load balancing is desired must be configured to allow load balancing actions and reporting, by setting attribute `EUtranCellRelation.loadBalancing`.

Load balancing between cells in different nodes requires an X2 connection between the nodes.

### System Functions

Table 2 System Function Dependencies

| Function | Description |
|---|---|
| Inter-frequency measurements | The number of inter-frequency measurements increases when the Inter-Frequency Load Balancing feature is activated.<br>The increase depends on the number of inter-frequency handovers required to keep traffic load in balance. |
| Inter-frequency handover | The number of inter-frequency handovers increases when the Inter-Frequency Load Balancing feature is activated.<br>The increase depends on the number of inter-frequency handovers required to keep traffic load in balance. |
| X2 signaling | X2 signaling is required for the exchange of traffic load information if the Inter-Frequency Load Balancing feature is applied between cells located on separate nodes. |
| Added E-RAB Establishment Success Rate | When the Inter-Frequency Load Balancing feature is activated, the accessibility success rate can slightly decrease for end-user services. This is carried by E-RABs included in the E-RAB setup procedure. |

### EPC Feature Dependencies

Packet Core Network-controlled RAN Load Balancing based on Mobility-Based Policy and RAN feature Inter-Frequency Load Balancing may have interoperability issues. This can cause unwanted behavior in the RAN or Packet Core Network, or both. See *Mobility-Based Policy* in the SGSN-MME CPI documentation for more information.

## 2.1 Compatibility of Load Balancing and Automatic Access Barring

> During ongoing automatic access class barring calculation mechanism, inter-frequency load balancing for incoming handover is disabled in the cell.

When automatic access class barring is used together with Inter-Frequency Load Balancing, the incoming handover to an overloaded cell is disabled.

Inter-frequency load balancing is enabled in the cell when the barring probability reaches `0`.

If load balancing during access class barring is enabled, that is `EUtranCellFDD.iflbAbEnabled` or `EUtranCellTDD.iflbAbEnabled` is set to `true`, the following events occur:

- The subscription ratio measurements are still ongoing in the cell when the automatic access class barring calculation mechanism is triggered.
- The function enables the cell for outgoing load balancing handover, and a faster recovery from overload.
- Triggering of the access class barring calculation mechanism does not disable subscription ratio evaluation and Inter-Frequency Load Balancing.

If the Preferential Traffic Management solution is enabled, by setting the `PtmFunction.ptmEnabled` attribute to `true`, and the cell is configured as priority cell, triggering of the access class barring calculation mechanism does not disable the following events:

- Subscription ratio evaluation
- Inter-Frequency Load Balancing
- Inter-Frequency Offload

A cell is configured as priority cell if the following is true:

- `PtmCellProfile.cellType` is set to `PRIORITY`.
- The `EUtranCellFDD.ptmCellProfileRef` or the `EUtranCellTDD.ptmCellProfileRef` reference to a cell profile is set as priority.

Example 1 Automatic Access Class Barring Disables Load Balancing

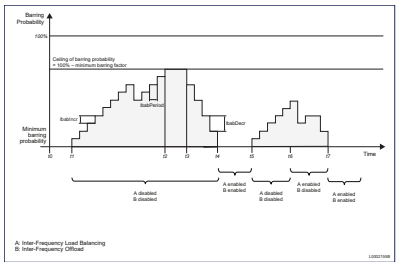



Figure 2 Inter-Frequency Load Management Disabled by Automatic Access Barring

The following events occur during the access class barring calculation mechanism:

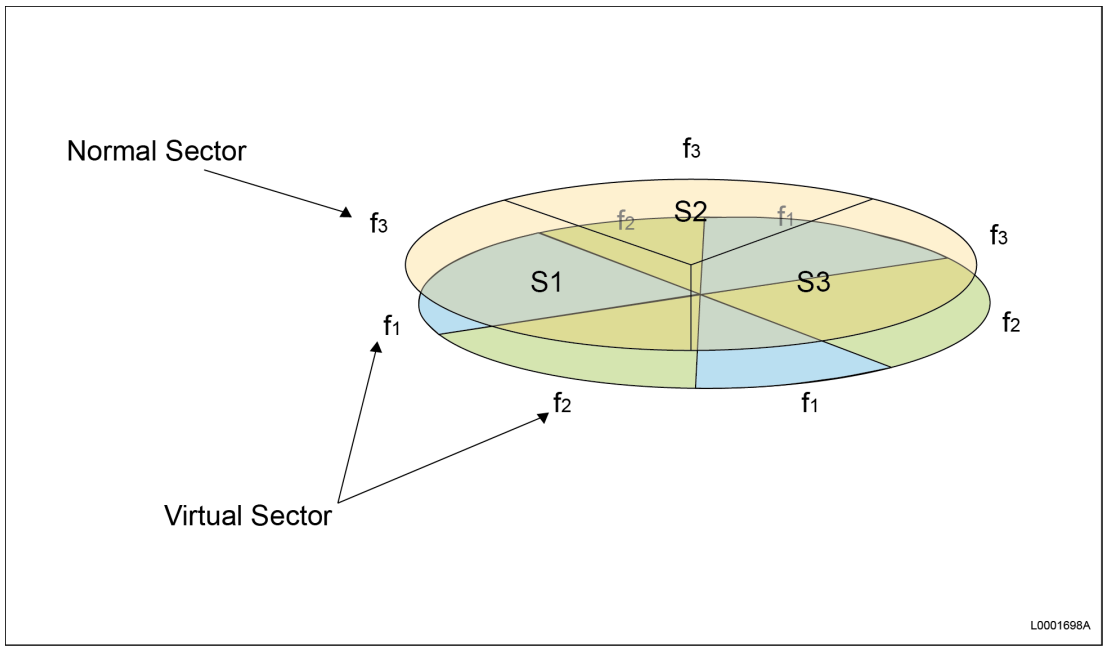
1. Before t1, inter-frequency load balancing can be enabled.
2. At t1, these load balancing functions are disabled, and automatic access class barring calculation is enabled.

3. Between t1 and t4, until the automatic access class barring calculation is enabled, the load balancing functions remain disabled.
4. At t4, the barring probability reaches its minimum value, that is `0`. Inter-frequency load balancing can be enabled again.
5. Between t5 and t6, inter-frequency load balancing is disabled again.
6. Between t6 and t7, outgoing inter-frequency load balancing is enabled only if `EUtranCellFDD.iflbAbEnabled` or `EUtranCellTDD.iflbAbEnabled` is set to `true`.
7. At t7, the barring probability reaches its minimum value again. Inter-frequency load balancing and inter-frequency offload are enabled again.

## 2.2 Compatibility of Inter-Frequency Load Balancing and Virtual Sectors

In the following cases there are compatibility issues between features Inter-Frequency Load Balancing and Virtual Sectors:

- When Inter-Frequency Load Balancing is activated after Virtual Sectors, the SCell candidate relations must be set to the overlapping virtual sectors.
- When Inter-Frequency Load Balancing is activated before Virtual Sectors, Inter-Frequency Load Balancing cannot work. However the Inter-Frequency Load Balancing configuration between these virtual sectors can be kept. Turn off the license for Virtual Sectors to use Inter-Frequency Load Balancing again.
- Inter-Frequency Load Balancing cannot work between virtual sectors. However Inter-Frequency Load Balancing is still possible between a virtual sector and a normal sector.




There are three carriers per radio. One carrier (f3) covers the full physical sector and the other two carriers (f1 and f2) cover the virtual sectors.

Figure 3 Compatible Configuration for Carrier Aggregation or Inter-Frequency Load Balancing

# 3 Feature Operation of Inter-Frequency Load Balancing

The Inter-Frequency Load Balancing feature is a continuous process that tries to balance the traffic load between carrier frequencies. Load balancing relations are configured between overlapping inter-frequency cells. The cells communicate information about traffic load and use it to regulate the load balancing algorithm. This allows the feature to operate autonomously in each cell without a central node. This is beneficial to the scalability of the feature.

## Feature Operation Sequence Diagram

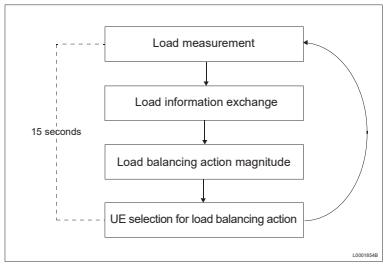



Figure 4 Load Balancing Operation

## Process Steps

1. The traffic load in a cell is assessed repeatedly in certain intervals of a load balancing cycle.

   The current subscription ratio of the cell is calculated. The QCI subscription quanta values (`QciProfilePredefined.qciSubscriptionQuanta` and `QciProfileOperatorDefined.qciSubscriptionQuanta`) of all E-RABs (both GBR and non-GBR services) present in the cell are added. The sum is divided by the cell subscription capacity that is the value of `EUtranCellFDD.cellSubscriptionCapacity` or `EUtranCellTDD.cellSubscriptionCapacity` attributes.

   The subscription ratio is input to the load information exchange and used to determine further load balancing actions in the cell.

2. Cell load information is exchanged between cells. Each cell can receive load reports from one or more other cells in the same node or in a remote node. The exchange of load information only takes place if load balancing is configured between the related cells.

   Load information exchange between cells in remote nodes is performed over X2 private messages, in an asynchronous process.

   Information exchange is only possible between Ericsson nodes.

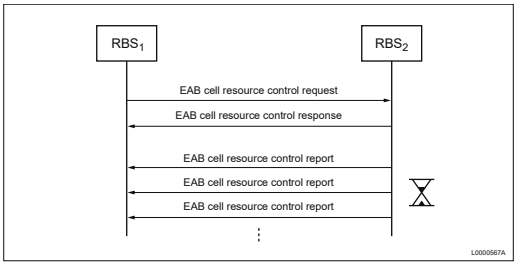



Figure 5 Load Information Exchange between Nodes

The selection of target cells is controlled with parameter EUtranCellRelation.loadBalancing.

3. Based on the load information received from other cells, each source cell determines the traffic load that is to be handed over to each target cell to balance the load differences.

   This is done based on the following:

   – Load difference to each target cell

   – Minimum load difference to trigger load balancing action, set by LoadBalancingFunction.lbThreshold

   – Total offload from the source cell

   – Maximum load difference, LoadBalancingFunction.lbCeiling, taken into account in one round of load-balancing action

   If multiple target cells exist, the total offload is split between these cells.

   Normally, load balancing targets equal subscription ratios in the source and target cells. However, when traffic is low, load balancing can allow rate offset. The rate offset is pronounced at very low subscription ratios and gradually decreases as traffic grows. The rate offset is removed at a subscription ratio of LoadBalancingFunction.lbRateOffsetLoadThreshold or higher, where load balancing targets equal subscription ratio.

4. UEs that are suitable for offload are selected.

   When uplink-heavy UE detection for the Inter-Frequency Load Balancing feature is configured and a UE is marked as uplink-heavy, it is excluded from load-balancing mobility actions. For more information, see Uplink-Heavy UE Detection.

   In the UE selection process, the UEs perform inter-frequency Event A5 measurements on the target frequency. This ensures that the UE has acceptable coverage in a target cell. Only those UEs are selected which support the target frequency band and trigger an Event A5 measurement report on the target cell.

   **Note:** Event A5 is also enables load balancing actions for UEs that are configured for Carrier Aggregation.

   The maximum number of measurements started for load balancing purposes during a 15-second load balancing cycle is limited.

   **Note:** A UE in connected mode is selected to perform a measurement on a target frequency for load balancing purposes at most once for each load balancing cycle.

5. The execution of load balancing actions is performed by handover of the selected UEs to the target cells.

   The handover of UEs is performed using the following normal inter-frequency handover procedures:

   – X2 Handover Preparation

   – S1 Handover Preparation

   – RRC Connection Reconfiguration

   Handover to a target cell is inhibited in the following cases:

   – The `EUtranCellRelation.isHoAllowed` attribute for the target cell is set to FALSE.

   – The cell is listed as forbidden in a HRL provided for the UE. For more information on the support of HRL, see Shared LTE RAN.

   – The RFPM function of the Subscriber Triggered Mobility feature is used. The `FreqPrioEUTRA.loadBalancingAllowed` attribute related to the `RATFreqPrio` MO for the target carrier frequency is set to FALSE for matching UE SPID.
     **Note:** If the Preferential Traffic Management solution is used, to inhibit inter-frequency offload for load balancing relations is inhibited. It can be achieved by setting the `EUtranCellRelation.loadBalancing` attribute to `IFO_AND_IFLB`, and the `FreqPrioEUTRA.offloadAllowed` attribute must also be set to FALSE.

     Otherwise, if the `FreqPrioEUTRA.loadBalancingAllowed` attribute is set to TRUE or there is no matching UE SPID, load balancing to the target cell for the UE is allowed.

     For more information on the support of SPID use, see Subscriber Triggered Mobility.

   If the solution Preferential Traffic Management is enabled and appropriate configuration is set, the following applies:A UE in connected mode is selected to perform a measurement on a target frequency for load balancing purposes at most once for each load balancing cycle.

   Inter-frequency load balancing can be stopped for some cells—to priority cells with high traffic load from non-priority cells. This allows the priority band to be used primarily to serve priority users, and ensures access for priority UEs in high traffic load situations.

   For more information, see Preferential Traffic Management Guidelines.

## 3.1 Load Balancing Cycle

> The load balancing cycle is the time interval between two successive evaluations of traffic load balancing status. It is the time frame for the actions attempting to restore the traffic load balance.

Load balancing between cells is done repeatedly in successive load balancing cycles of 15 seconds.

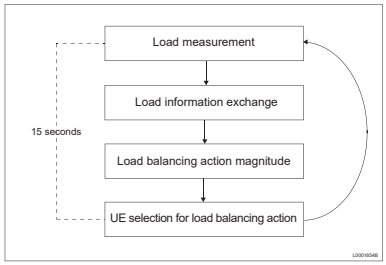



Figure 6 Load Balancing Cycle

| | |
|---|---|
| Load Measurement | The load of each cell is estimated. |
| Load Information Exchange | Load information is exchanged between source and target cells.<br>For inter-eNodeB load exchange, X2 messages are used. |
| Load Balancing Action Magnitude | The amount of offloading required to manage load between cells is determined.<br>Higher loaded cells select some UEs as candidates for load balancing. The selected UEs are configured for event A5 inter-frequency measurement. |
| UE Selection for Load Balancing Action | The UEs suitable for offload are selected. The suitable UEs are those that reported to the source cell that they detected sufficient coverage in a target cell.<br>Some suitable UEs are moved to detected target cells by inter-frequency handover. |

## 3.2 Load Balancing Relation

> The load balancing relation is the logical relation between a load balancing source and a target cell. Load balancing relations are controlled with the `EUtranCellRelation.loadBalancing` attribute.

For load balancing features to steer traffic from one source cell to a target cell, a relation must be defined in the source cell. This relation is defined to the target cell using the `EUtranCellRelation` MO. The `EUtranCellRelation.loadBalancing` attribute controls whether load balancing, or offload, or both are allowed.

Note: Each defined load balancing relation is unidirectional.

## 3.3 User Experience Dependency on Traffic Load

The influence of traffic load on the user experience is different for GBR services with guaranteed performance and non-GBR services of best effort type.

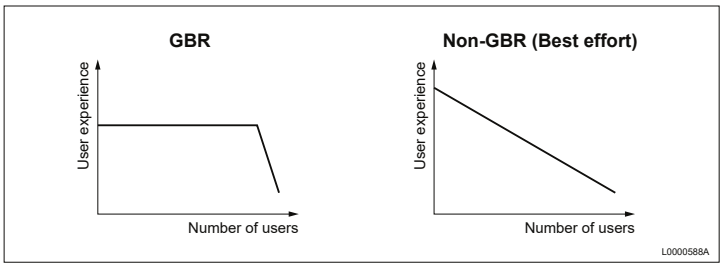


Figure 7 Influence of Number of Users on User Experience for GBR and Non-GBR Services

The user experience of a GBR service is essentially unaffected by an increase of traffic load. This is up to the point where LTE RAN is unable to grant additional users into the network. Above that point, LTE RAN may deny access and the user experience drops rapidly for users with difficulties to gain access.

The user experience of non-GBR services gradually decreases by an increase of traffic load. The grade of service is characterized by qualities like achievable bit rate and packet delay. The achievable bit rate for the individual user decreases and packet delay tends to increase. This is because shared transmission resources are used in LTE RAN with an increasing traffic load.

The LTE RAN provides both GBR and non-GBR services. Of the total transmission resource in the network a certain amount is employed to satisfy the GBR services. This reduces the resources available for non-GBR services. Therefore, the number of both GBR and non-GBR users affects the non-GBR user experience. The GBR user experience, on the other hand, is to a high degree unaffected by the number of non-GBR users.

**Note:** Admission thresholds for GBR services may be used to ensure a minimum resource available for non-GBR services. For this, the Dynamic GBR Admission Control feature can be used.

## 3.4 GBR Traffic Load

GBR traffic is characterized by a relatively persistent E-RAB that generates a steady data stream. For example, an E-RAB for a voice call typically lasts for one or a few minutes. During the voice call, a stream of voice packets is transferred at a nearly constant rate.

Therefore, the number of E-RABs for GBR traffic present in a cell is a good basis for the assessment of the GBR traffic load.

## 3.5 Non-GBR Traffic Load

Non-GBR traffic is typically characterized by short but frequent data transactions. The individual data transactions can range from a few data packets to the transfer of large files. Many data transactions are completed within a short time, often less than a few seconds.

Non-GBR traffic has the following pattern:

1. The UE enters connected mode.
2. The E-RABs are established.
3. The data transaction, or a sequence of data transactions proceeds.
4. The UE is released after a period of inactivity.

This pattern is repeated, but each RRC connection and each E-RAB for the node is unique. The node is unable to correlate the different RRC connections generated by a UE.

Non-GBR traffic patterns are difficult to predict because they change often and quickly. The activity of an individual UE or E-RAB does not necessarily give much information about the activity a few seconds ahead. The number of E-RABs presented by the UEs to the LTE RAN can be used to assess the non-GBR traffic load generated by the UE groups. This applies for a UE group camped in a cell.

## 3.6 Subscription Ratio

> Subscription ratio is a relative traffic load measure. It is the ratio between the total QCI subscription quanta of all E-RAB instances currently configured in a cell and the cell subscription capacity of the cell.

It takes a value between zero and one for an under-subscribed cell. Values greater than one indicate an over-subscribed cell.

$$SubscriptionRatio = \frac{\sum qciSubscriptionQuanta}{cellSubscriptionCapacity}$$

Figure 8 Subscription Ratio

The configuration of the following values has to obtain the required load balancing behavior:

- `qciSubscriptionQuanta` (in MOs `QciProfilePredefined` or `QciProfileOperatorDefined`)
- `cellSubscriptionCapacity` (in MOs `EUtranCellFDD` or `EUtranCellTDD`)

This enables load balancing between cells with different UE groups, and cells with different traffic capacities.

Configuration of the assessment model offers control of the load balancing objectives in LTE RAN and opportunity for further changes.

The subscription ratio can be used to monitor the traffic load in a cell.

## 3.7 Cell Subscription Capacity

Cell subscription capacity is the nominal traffic handling capacity of a cell. Its value is an estimate of the total amount of QCI subscription quanta values the cell is expected to accommodate at full system load. Cell subscription capacity defines the target for load balancing to divide the traffic load between overlaid cells within the network. It can be configured with one of the following MOM attribute:

- `EUtranCellFDD.cellSubscriptionCapacity`
- `EUtranCellTDD.cellSubscriptionCapacity`

## 3.8 QCI Subscription Quanta

QCI subscription quanta is the representation of the anticipated or typical traffic handling capacity required for an E-RAB with certain QCI.

QCI subscription quanta is configured for a QoS class by setting the `QciProfilePredefined.qciSubscriptionQuanta` or `QciProfileOperatorDefined.qciSubscriptionQuanta` attribute.

## 3.9 Rate Offset

Rate offset is traffic load balancing offset, shifting the traffic load with a bias towards cells with high traffic handling capacity. This offers anticipated better performance during periods of low traffic.

Normally, load balancing targets equal subscription ratios in the source and target cells. However, when traffic is low, load balancing can allow higher subscription ratios in high capacity cells, as those provide better user performance in sparsely loaded LTE RAN. The effect is called rate offset. The rate offset is pronounced at very low subscription ratios and gradually decreases as traffic grows. The rate offset is completely removed at a subscription ratio of `LoadBalancingFunction.lbRateOffsetLoadThreshold` and higher, where load balancing targets equal subscription ratio.

## 3.10 Load Management and MP Load Control

The objective for the load management functions in the eNodeB is to balance or offload load when possible. However, the focus is on the air interface load. During the execution of load management functions the MP processor load may increase. To not affect general traffic handling, the load management functions are temporarily paused if the MP load control state is higher than `NOT HIGH`. For more information, see Basic Load Management.

## 3.11 Sticky Carrier

The Sticky Carrier configuration is used to confine UEs in idle mode to the carrier frequency of the current serving cell. This is suitable in a network deployment where good coverage can be expected across large areas on two or more carrier frequencies.

The technique is shown in Table 3. It takes advantage of the CRPs of the carrier frequencies. The CRP settings are included in the System Information Broadcast in each cell. In the Sticky Carrier configuration, the CRP of the serving carrier frequency is set higher than the CRP of the other carrier frequencies used in the network.

In this way, a UE in idle mode is confined to the carrier frequency of the current serving cell. This is beneficial because the UE tends to remain on the carrier frequency where it is released to idle mode. The effect of load balancing actions in connected mode may then sustain periods in idle mode. The effect remains the next time the UE connects to the network. It reduces the need for load balancing action to compensate for idle mode mobility between the carrier frequencies.

Table 3 Example of Cell Reselection Priority with Sticky Carrier

| Serving Carrier Frequency | CRP Broadcast in Cells on FreqA | CRP Broadcast in Cells on FreqB | CRP Broadcast in Cells on FreqC |
|---|---|---|---|
| FreqA | 2 | 1 | 1 |

| Serving Carrier Frequency | CRP Broadcast in Cells on FreqA | CRP Broadcast in Cells on FreqB | CRP Broadcast in Cells on FreqC |
|---|---|---|---|
| FreqB | 1 | 2 | 1 |
| FreqC | 1 | 1 | 2 |

## 3.12 Priority Carrier

The Priority Carrier configuration is used to direct UEs in idle mode to a particular carrier frequency. It is suitable for deployment in networks where the cells on one or more carrier frequencies are small and do not provide continuous coverage. In those cells, UE camping frequently runs into poor coverage. As a result, UEs are required to reselect cells on other carrier frequencies. This configuration supports the Inter-Frequency Load Balancing feature to find a sufficient number of UEs in connected mode suitable for relocation back to the priority carrier cells. Using the idle mode mobility to bias the UE population towards the priority carrier can then be a useful complement.

The technique of Priority Carrier is shown in Table 4. The carrier frequency C is configured as the priority carrier, whereas a sticky carrier configuration is applied between carrier frequencies A and B. Strictly higher CRP is configured for C, the priority carrier frequency, compared with A and B, the other carrier frequencies.

Table 4 Example of Cell Reselection Priority with Priority Carrier

| Serving Carrier Frequency | CRP Broadcast in Cells on FreqA | CRP Broadcast in Cells on FreqB | CRP Broadcast in Cells on FreqC |
|---|---|---|---|
| FreqA | 2 | 1 | 1 |
| FreqB | 1 | 2 | 1 |
| FreqC | 3 | 3 | 3 |

UEs in idle mode are directed to the priority carrier cells when entering their coverage. The Inter-Frequency Load Balancing feature then has an easier task to find suitable UEs in connected mode for load balancing action towards those cells. This can be beneficial, in particular if the priority carrier cells only cover a small number of the UEs on the other carrier frequencies.

## 3.13 Uplink-Heavy UE Detection

When uplink-heavy UE detection for the feature is configured, the UE selection process improves for load-balancing mobility actions. The feature excludes UEs from load-balancing mobility actions if they are marked as uplink-heavy. The uplink-heavy UE detection mechanism is applied to all UEs that connect to the LTE cell.

The information about the uplink-heavy state of the UE is stored in the UE context.

# 4 Network Impact of Inter-Frequency Load Balancing

### Capacity and Performance

The Inter-Frequency Load Balancing feature adjusts uneven distribution of traffic load between carrier frequencies. The ability to fully use the traffic capacity on all the available carrier frequencies is improved. The feature is beneficial for the system throughput and the number of UEs and E-RABs the LTE RAN is able to handle. It allows the system capacity to increase proportionally when new carrier frequencies are added and the available RF spectrum thereby increases.

The feature relies on UE inter-frequency measurements to identify UEs suitable for load balancing action. The RRC signaling associated with inter-frequency measurements and the handover-related signaling may increase. At steady state, when traffic load is nearly in balance, the increase is small compared with other signaling of similar kind.

### Mobility

The Inter-Frequency Load Balancing feature redistributes users in connected mode between inter-frequency cells by means of traffic load triggered inter-frequency handover. The number of inter-frequency handover increases with this feature. The handover is typically performed in good radio conditions.

The outgoing handover cause code value for the Inter-Frequency Load Balancing feature is set depending on the value of the attributes `LoadBalancingFunction.lbCauseCodeS1SourceTriggersOffload` and `LoadBalancingFunction.lbCauseCodeX2SourceTriggersOffload` for S1 and X2 handovers respectively. It is important that Inter-Frequency Load Balancing and Inter-Frequency Offload use different cause codes. Therefore the selected cause code for Inter-Frequency Load Balancing is set by the node depending on the value of the parameters.

Table 5 Cause Code for Outgoing S1 Handover

| lbCauseCodeS1SourceTriggersOffload | Cause Code |
|---|---|
| REDUCE_LOAD_IN_SERVING_CELL | Resource Optimization Handover |
| RESOURCE_OPTIMISATION_HANDOVER | Reduce Load In Serving Cell |

Table 6 Cause code for Outgoing X2 Handover

| lbCauseCodeX2SourceTriggersOffload | Cause Code |
|---|---|
| REDUCE_LOAD_IN_SERVING_CELL | Resource Optimization Handover |
| RESOURCE_OPTIMISATION_HANDOVER | Reduce Load In Serving Cell |
| LOAD_BALANCING | Resource Optimization Handover |

The following sub-counters are introduced to distinguish inter-frequency load balancing handover from other types of inter-frequency handover:

- pmHoPrepAttLteInterFLb
- pmHoPrepSuccLteInterFLb
- pmHoExeSuccLteInterFLb

# 5 Parameters for Inter-Frequency Load Balancing

### Feature Configuration Parameters

- EUtranCellFDD.cellSubscriptionCapacity
- EUtranCellTDD.cellSubscriptionCapacity
- EUtranFreqRelation.lbA5Thr1RsrpFreqOffset
- EUtranCellRelation.loadBalancing
- LoadBalancingFunction.lbCeiling
- LoadBalancingFunction.lbRateOffsetCoefficient
- LoadBalancingFunction.lbRateOffsetLoadThreshold
- LoadBalancingFunction.lbThreshold
- QciProfileOperatorDefined.qciSubscriptionQuanta
- QciProfilePredefined.qciSubscriptionQuanta
- ReportConfigEUtraInterFreqLb.a5Threshold1Rsrp
- ReportConfigEUtraInterFreqLb.a5Threshold2Rsrp
- ReportConfigEUtraInterFreqLb.a5Threshold2Rsrq
- ReportConfigEUtraInterFreqLb.hysteresisA5

### Affected Parameters

Table 7 Affected Parameters

| Parameter | Description |
|---|---|
| UeMeasControl.sMeasure | Serving cell RSRP threshold controlling whether the UE is required to perform measurements<br>The parameter should be disabled in cells where load balancing is applied. |

### Parameters Affecting the Feature

Table 8 Parameters Affecting the Feature

| Parameter | Description |
|---|---|
| ENodeBFunction.ulHeavyUeDetectTime | The attribute can be used to configure uplink-heavy UE detection. |
| ENodeBFunction.ulHeavyUeBsrThresh | The attribute can be used to configure uplink-heavy UE detection. |
| ENodeBFunction.ulHeavyUeBsrHyst | The attribute can be used to configure uplink-heavy UE detection. |
| EUtranCellFDD.iflbAbEnabled | If the EUtranCellFDD.iflbAbEnabled or the EUtranCellTDD.iflbAbEnabled parameter is set to true, the Inter-Fequency Load Balancing feature is enabled during access class barring. |
| EUtranCellTDD.iflbAbEnabled | |
| loadBalancingAllowed | Member of the FreqPrioEUTRA struct. The parameter references from the RATFreqPrio MO.<br>If the RFPM function of Subscriber Triggered Mobility is activated and the UE has a SPID value associated with it, the inter-frequency load balancing is inhibited for that UE. It is inhibited only when the loadBalancingAllowed attribute related to the concerned RATFreqPrio MO for the concerned target carrier frequency is set to false. |
| EUtranCellFDD.lbUlHeavyUeDetectEnabled | The attribute can be used to enable uplink-heavy UE detection for load-balancing mobility actions. |
| EUtranCellTDD.lbUlHeavyUeDetectEnabled | The attribute can be used to enable uplink-heavy UE detection for load-balancing mobility actions. |
| PtmFunction.ptmEnabled | These parameters are used to enable and configure the Preferential Traffic Management solution. With appropriate configuration, inter-frequency load balancing can be stopped for some cell relations—to priority cells with high traffic load from non-prioritized cells. For more information, see Preferential Traffic Management Guidelines. |
| PtmIflbConfig.stopIncomingIflbThreshold | |
| PtmCellProfile.cellType | |
| PtmIflbConfig.stopOutgoingIflbEnabled | |

# 6 Performance of Inter-Frequency Load Balancing

## KPIs

The table lists the main KPIs associated with the feature, and the expected impact.

Table 9 Key Performance Indicators

| KPI | Description |
|---|---|
| Mobility Success Rate | The Mobility Success Rate KPI includes both the preparation of the target cell resources and the move from the source cell to the target cell. |

## Counters

The following counters are associated with the Inter-Frequency Load Balancing feature:

- EUtranCellFDD.pmLbSubRatioSamp
- EUtranCellTDD.pmLbSubRatioSamp
- EUtranCellFDD.pmLbSubRatioSum
- EUtranCellTDD.pmLbSubRatioSum
- EUtranCellRelation.pmLbMeasRepUe
- UtranCellRelation.pmLbMeasRepUe
- EUtranCellRelation.pmLbQualifiedUe
- UtranCellRelation.pmLbQualifiedUe
- EUtranCellRelation.pmHoPrepAttLteInterFLb
- EUtranCellRelation.pmHoPrepSuccLteInterFLb
- EUtranCellRelation.pmHoExeAttLteInterFLb
- EUtranCellRelation.pmHoExeSuccLteInterFLb
- EUtranCellRelation.pmHoPrepAttNonMob
- UtranCellRelation.pmHoPrepAttNonMob
- EUtranCellRelation.pmHoPrepSuccNonMob
- UtranCellRelation.pmHoPrepSuccNonMob
- EUtranCellRelation.pmHoExeAttNonMob
- UtranCellRelation.pmHoExeAttNonMob
- EUtranCellRelation.pmHoExeSuccNonMob
- UtranCellRelation.pmHoExeSuccNonMob
- EUtranCellFDD.pmPtmStopIncomingIflb
- UtranCellRelation.pmPtmStopIncomingIflb
- EUtranCellFDD.pmPtmSubRatioSumDistr
- EUtranCellTDD.pmPtmSubRatioSumDistr

More information about counters can be found in Managed Object Model (MOM).

## Events

Table 10 Events

| Event | Event Parameter | Description |
|---|---|---|
| INTERNAL_EVENT_LB_SUB_RATIO | EVENT_PARAM_SUB_RATIO | Average subscription ratio for a cell during a load reporting period. Default time is 15 sec. |
| X2_PRIVATE_MESSAGE | ASN.1 encoded content is part of the pmEventHeader | X2 messages used to subscribe and exchange load information between RBSs. |
| INTERNAL_EVENT_LB_INTER_FREQ | EVENT_PARAM_NEIGHBOR_CGI | CGI of target cell that participates in load balancing if it is available. |

| Event | Event Parameter | Description |
|---|---|---|
| | EVENT_PARAM_LB_DIFF | Load difference (in subscription ratio) between source and neighbor cells. |
| | EVENT_PARAM_LB_DIFF_INT_SIGN | Unsigned integer value of EVENT_PARAM_LB_DIFF. The value can be positive or negative. |
| | EVENT_PARAM_LB_AMOUNT | Load balancing amount (in subscription quanta) to be moved from source to target cell. |
| | EVENT_PARAM_LB_SOURCE_RATIO | Subscription ratio of source cell. |
| | EVENT_PARAM_LB_TARGET_RATIO | Subscription ratio of target cell. |
| | EVENT_PARAM_LB_RATE_OFFSET | Rate offset applied to offset the load balancing towards a cell with higher capacity, if the general traffic load level is low. |
| | EVENT_PARAM_LB_OFFSET_INT_SIGN | Unsigned integer value of EVENT_PARAM_LB_RATE_OFFSET. The value can be positive or negative. |
| INTERNAL_EVENT_UE_LB_MEAS | EVENT_PARAM_FREQ | Target frequency |
| | EVENT_PARAM_TRIGGERING_FUNCTION | Indicates the function that has triggered UE measurement. Can be one of the following:<br>– Load Balancing<br>– Carrier Aggregation Triggered Redirection (CATR)<br>– Best Neighbor Relation (BNR) for Load Balancing<br>– Admission-Triggered Offload (ATO) |
| | EVENT_PARAM_UE_CA_CABABLE | Indicates whether or not the UE selected for load balancing related measurements is CA-capable. |
| | EVENT_PARAM_NEIGHBOR_CGI | CGI of neighbor target cells for which UEs fulfilled event A5 criteria. |
| | EVENT_PARAM_TRAVERSED_UES | Number of UEs checked and discarded before choosing this UE for load balancing related measurements. |
| INTERNAL_EVENT_UE_LB_QUAL | EVENT_PARAM_NEIGHBOR_CGI | CGI of neighbor target cells for which UEs fulfilled event A5 criteria. |
| | EVENT_PARAM_HO_SOURCE_OR_TARGET_TYPE | Indicates the RAT of the source for incoming handover and for target for outgoing handover cell. |
| | EVENT_PARAM_LB_CANCELLED_REASON | Indicates the reason for not proceeding with a load balancing (LB) or CATR when a measurement report has been received from the UE :<br>– CA criteria not met<br>– Reported cell is not a valid load balancing target<br>– Reported cell is a valid target, but lbAmount is already satisfied for the cell<br>– Service Specific Load Management (SSLM) related restrictions<br>– RSRQ of target cell is lower than configured threshold<br>– UE has signalled interest in Multimedia Broadcast Multicast Service (MBMS) on serving frequency<br>– UE has signalled interest in Multimedia Broadcast Multicast Service (MBMS) and UE is MBMS on SCell capable ( mbms-SCell-r11 = 1, 3GPP 36.331)<br>– UE has an ongoing Voice over IP (VoIP) call and does not support Single Radio Voice Call Continuity (SRVCC)<br>– Admission-Triggered Offload (ATO) related restrictions |
| | EVENT_PARAM_TRIGGERING_FUNCTION | Indicates the function that has triggered the UE measurement.<br>Can be one of the following:<br>– Load Balancing<br>– CATR<br>– BNR for Load Balancing<br>– Admission-Triggered Offload (ATO) |

| Event | Event Parameter | Description |
|---|---|---|
| INTERNAL_PROC_HO_EXEC_X2_OUT | EVENT_PARAM_PROC_HO_EXEC_OUT_ATTEMPT_CAUSE | The cause of the procedure for HO Execution.<br>– Circuit Switched (CS)<br>– Carrier Aggregation (CA)<br>– Service or Priority-triggered Inter-Frequency Handover (SPIFHO)<br>– Admission-Triggered Offload (ATO) |
| INTERNAL_PROC_HO_EXEC_S1_OUT | EVENT_PARAM_PROC_HO_EXEC_OUT_ATTEMPT_CAUSE | The cause of the procedure for HO Execution.<br>– Circuit Switched (CS)<br>– Carrier Aggregation (CA)<br>– Service or Priority-triggered Inter-Frequency Handover (SPIFHO)<br>– Admission-Triggered Offload (ATO) |
| INTERNAL_EVENT_UE_MOBILITY_EVAL | EVENT_PARAM_MOBILITY_TRIGGER | Indicates the trigger for performing a mobility evaluation.<br>– Circuit Switched (CS)<br>– Carrier Aggregation (CA)<br>– Service or Priority-triggered Inter-Frequency Handover (SPIFHO)<br>– Admission-Triggered Offload (ATO ) |

# 7 Activate Inter-Frequency Load Balancing

> To use the feature, it must be activated following the usual feature activation procedure.

### Prerequisites

- The license key is installed in the node.
- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.
- The Coverage-Triggered Inter-Frequency Handover feature is activated.

### Steps

1. Set the `FeatureState.featureState` attribute to `ACTIVATED` in the `FeatureState=CXC4011319` MO instance.

### After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

# 8 Deactivate Inter-Frequency Load Balancing

> If the feature is no longer needed, it can be deactivated following the usual feature deactivation procedure. It must also be deactivated before the activation of any conflicting feature.

### Prerequisites

CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

### Steps

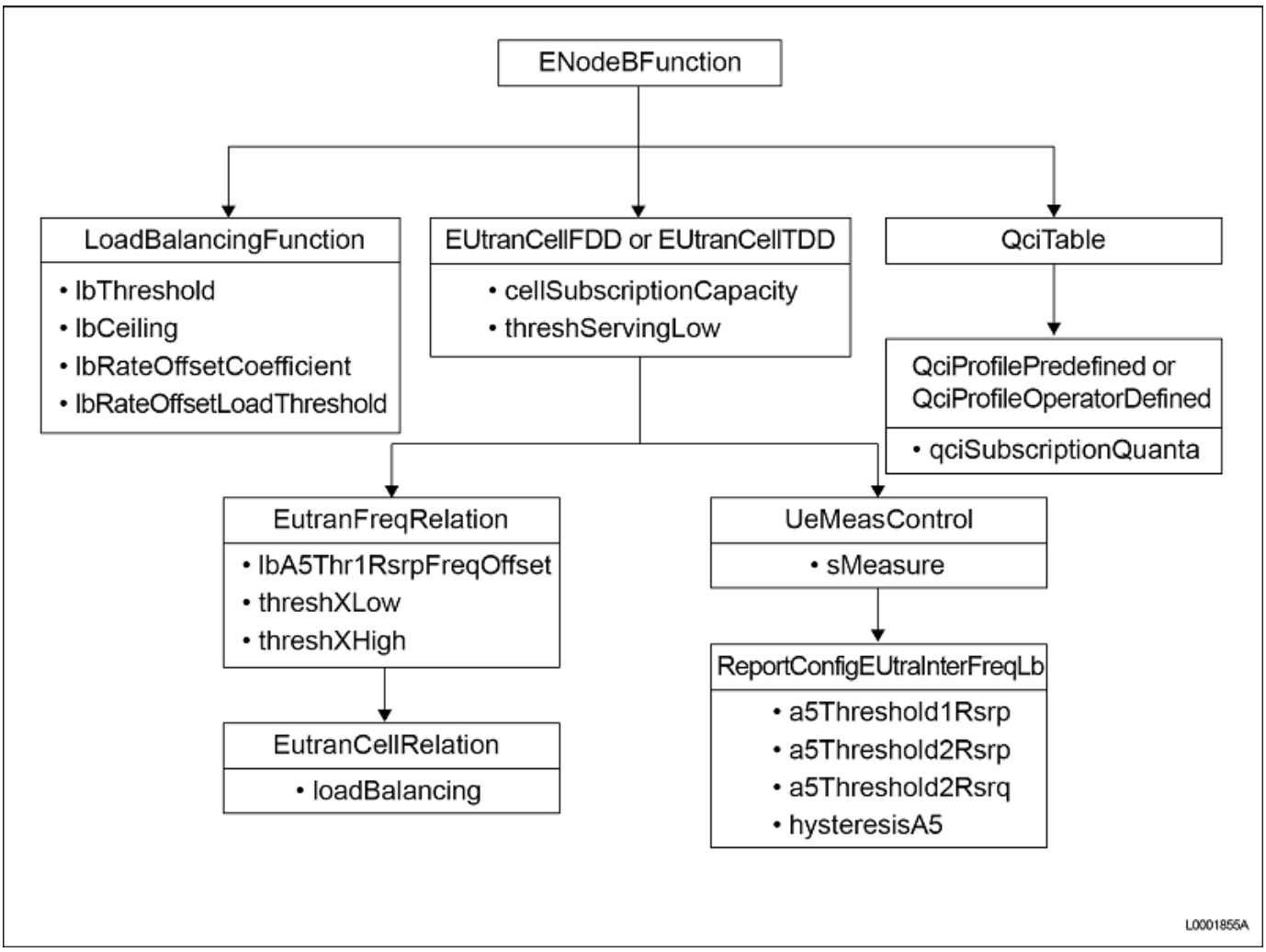
1. Set the `FeatureState.featureState` attribute to `DEACTIVATED` in the `FeatureState=CXC4011319` MO instance.

### After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

# 9 Engineering Guidelines for Inter-Frequency Load Balancing

For information on configuration, click on the attribute.




## 9.1 Configuration of Load Balancing Relation

> Load balancing relations are configured in the source cell towards each target cell, by setting the `EUtranCellRelation.loadBalancing` attribute.

Table 11 Configuration of Load Balancing Relation

| MO Attribute | Value | Description |
|---|---|---|
| `EUtranCellRelation.loadBalancing` | `NOT_ALLOWED` | Setting the `EUtranCellRelation.loadBalancing` attribute to `NOT_ALLOWED` deactivates Inter-Frequency Load Balancing and Inter-Frequency Offload to the related target cell, in one direction. |
| | `ALLOWED` | Setting the `EUtranCellRelation.loadBalancing` attribute to `ALLOWED` activates Inter-Frequency Load Balancing to the related target cell, in one direction. |
| | `OFFLOAD` | Setting the `EUtranCellRelation.loadBalancing` attribute to `OFFLOAD` activates Inter-Frequency Offload to the related target cell, in one direction. |
| | `IFO_AND_IFLB` | Setting the `EUtranCellRelation.loadBalancing` attribute to `IFO_AND_IFLB` activates both Inter-Frequency Load Balancing and Inter-Frequency Offload to the related target cell, in one direction.<br>**Note:** The `IFO_AND_IFLB` value can only be used if Preferential Traffic Management is enabled and configured. |

The `EUtranCellRelation.loadBalancing` attribute can be configured in one direction, from the source cell to the target cell.

To apply the same configuration in both directions between two cells, `EUtranCellRelation.loadBalancing` must be set to the same value in both directions. This allows both cells to act as source cells to each other.

The `EUtranCellRelation.loadBalancing` attribute also controls the exchange of load information. When set to `ALLOWED` or `IFO_AND_IFLB`, the source cell requests the target cell to provide traffic load reports.

Other mobility functionality is not affected by the `EUtranCellRelation.loadBalancing` attribute.

Note: Inter-Frequency Load Balancing does not consider and may override restriction of connected mode mobility at coverage-triggered session continuity. Coverage-Triggered Inter-Frequency Session Continuity describes the configuration of the connected mode mobility restrictions.

The `EUtranCellRelation.loadBalancing` attribute can be determined and configured automatically by the Best Neighbor Relations for Intra-LTE Load Management feature.

**Related concepts**
Load Balancing Relation

## 9.2 Configuration of Subscription Parameters

The operation of the Inter-Frequency Load Balancing feature requires that QCI subscription quanta and cell subscription capacity are configured. QCI subscription quanta can be the `QciProfilePredefined.qciSubscriptionQuanta` or `QciProfileOperatorDefined.qciSubscriptionQuanta` attribute. The cell subscription capacity can be the `EUtranCellFDD.cellSubscriptionCapacity` or `EUtranCellTDD.cellSubscriptionCapacity` attribute.

### Configuration of QCI Subscription Quanta

The `QciProfilePredefined.qciSubscriptionQuanta` or `QciProfileOperatorDefined.qciSubscriptionQuanta` attributes are configured in proportion to the typical traffic load an E-RAB of the specific QoS class represents in LTE RAN at high load. The general recommendation is to choose values for these attributes in proportion to the expected downlink bit rates (in kbps) under those conditions.

Note: The `qciSubscriptionQuanta` attribute does not influence the scheduling of the E-RAB, nor is it used for admission control. It is used solely to express the typical traffic load an E-RAB of the specific QoS class imposes on LTE RAN.

In case of combination with the Automated Cell Capacity Estimation feature, the principle of configuring `qciSubscriptionQuanta` is changed. For more information, see the document Automated Cell Capacity Estimation.

For a GBR type of E-RAB, the `qciSubscriptionQuanta` parameter is chosen in proportion to the typical guaranteed downlink bit rate (in kbps) for the particular QCI.

Table 12 Recommendations to Configure qciSubscriptionQuanta for GBR E-RABs

| QCI | Recommendation |
|---|---|
| 1 (conversational voice) | The value depends on the typical voice codec in use. |
| Other (2..4, ...) | The value depends on the type of streaming media. |

For a non-GBR type of E-RAB, the `qciSubscriptionQuanta` parameter is chosen in proportion to what is considered an acceptable downlink bit rate (in kbps) for a UE in connected mode in LTE RAN at high load.

Table 13 Recommendations to Configure qciSubscriptionQuanta for Non-GBR E-RABs

| Type of E-RAB | Recommendation |
|---|---|
| Default non-GBR types | The default non-GBR types of E-RAB are activated whenever the UE enters connected mode, irrespective if they are going to be used. Conservative values are chosen, reflecting an expected low or moderate duty rate on these E-RABs.<br>In particular, the inactivity timer configured in the cell must be considered. A reduction of the inactivity timer may reduce the UE time spent in connected mode, whilst the UE data consumption is unaffected. The value of the `qciSubscriptionQuanta` attribute is increased to reflect the typical increase in average bit rate. If the inactivity timer is increased, the value of `qciSubscriptionQuanta` should be decreased.<br>Note: The document Radio Bearer Service describes the configuration of the inactivity timer. |
| IMS signaling (default QCI = 5) | The expected bit rate is typically low (not exceeding 3 kbps). |

| Type of E-RAB | Recommendation |
|---|---|
| Default ISP E-RAB | A suitable value can be based on the typical UE data consumption. This consumption is over a longer period of time in relation to the time spent in connected mode over the same period of time. The value has to focus on the conditions in LTE RAN at high load. |
| On-demand non-GBR type | Values have to be chosen based on the typical minimum QoS requirements of the intended applications. |

The default E-RAB can be distinguished on different QCIs for users of separate categories, for example, for users with different types of UE devices. It offers a way to differentiate the `qciSubscriptionQuanta` attribute values among users of different categories. This can be useful if there are user categories with widely different QoS demand.

Note: A separation of user categories on separate QCIs requires support from the core network. This is not a feature of LTE RAN.

### Configuration of Cell Subscription Capacity

At high load in LTE RAN, the `cellSubscriptionCapacity` (in MOs `EUtranCellFDD` or `EUtranCellTDD`) is configured in proportion to the achievable traffic capacity in the cell.

The achievable traffic capacity is typically proportional to the RF bandwidth of the cell, but can also depend on other factors, including the following:

- Inter-site distances and the presence of radio interference from neighbor cells.
- How the cell antenna is configured. For example, whether MIMO is applied.
- How the UE group in the cell is distributed between the cell center and the cell edge.

Note: If Inter-Frequency Load Balancing is used in combination with the features Automated Cell Capacity Estimation and Limited-Uplink-Aware IFLB, the function of the `cellSubscriptionCapacity` attribute can be replaced by these features. This determines and configures the cell capacity automatically. In this case, the principle of configuring `qciSubscriptionQuanta` is changed. For more information about the configuration, see the documents Automated Cell Capacity Estimation and Limited-Uplink-Aware IFLB.

It is recommended to choose a value of the attribute `cellSubscriptionCapacity` in proportion to the achievable downlink bit rate (in kbps) in the cell, considering the factors affecting the capacity. In a typical LTE RAN deployment, without advanced antenna configurations, the achievable spectrum efficiency at high load is usually in the proximity of 2 Mbit/MHz (bit rate for each RF bandwidth).

The amount of E-RABs the cell can accommodate is ultimately a trade-off with respect to the achievable QoS level. Increasing the amount of E-RABs has the following effects:

- Reduced throughput and increased packet delay for individual (non-GBR) E-RABs
- Increased subscription ratio, with a given configuration of the `qciSubscriptionQuanta` and `cellSubscriptionCapacity` attributes
- Declined user experience, yet a still operable cell

Therefore, the subscription ratio can be used as an indicator of the user experience in the cell.

The Inter-Frequency Load Balancing feature can handle load balancing also at subscription ratio levels greater than 1.0 (normally conceived as a mark of high load). The upper limit of the subscription ratio is 16. If the upper limit is reached, the load balancing can stall, because traffic load differences between cells above the limit are not perceptible. The value of the `cellSubscriptionCapacity` attribute should be chosen large enough to keep a margin with respect to this limitation.

**Related concepts**
Subscription Ratio
Cell Subscription Capacity
QCI Subscription Quanta
GBR Traffic Load
Non-GBR Traffic Load

## 9.3 Tuning of Subscription Parameters

In addition to the basic recommendations to configure load balancing, further tuning of the QCI subscription quanta and cell subscription capacity is possible to improve the load balancing behavior.

The QCI subscription quanta can be the `QciProfilePredefined.qciSubscriptionQuanta` or `QciProfileOperatorDefined.qciSubscriptionQuanta` MO attribute. The cell subscription capacity can be the `EUtranCellFDD.cellSubscriptionCapacity` or `EUtranCellTDD.cellSubscriptionCapacity` MO attribute.

For example, if a high E-RAB density of certain QoS class tends to cause the QoS level to decline. Then an increase of the corresponding `qciSubscriptionQuanta` attribute can help to reduce the number of UEs in the cells where that type of E-RAB is frequent.

The load balancing process does not consider the absolute level of `qciSubscriptionQuanta` and `cellSubscriptionCapacity`. The relations are important between these parameters.

### Relation Between qciSubscriptionQuanta Values for Different QCIs

The relations between `qciSubscriptionQuanta` values for different QCIs determine how UEs with different E-RAB configuration can be exchanged in the load balancing process.

Increasing the `qciSubscriptionQuanta` attribute value for one QoS class tends to relocate UEs from cells with high density of this type of E-RAB to cells with lower density. The opposite is also true.

### Relation Between cellSubscriptionCapacity Values of Different Cells

The relations between the `cellSubscriptionCapacity` values of different cells determine how the UE group is divided between cells where load balancing is allowed.

Increasing the value of the `cellSubscriptionCapacity` attribute for one cell attracts UEs to that cell. The opposite is also true.

### Relation Between cellSubscriptionCapacity and qciSubscriptionQuanta Values

The overall relation between the values of `cellSubscriptionCapacity` and `qciSubscriptionQuanta` attributes determines the absolute level of the subscription ratio in the cell.

Increasing the overall ratio between the values of attributes `cellSubscriptionCapacity` and `qciSubscriptionQuanta` requires a higher number of UEs in the cell to reach the same subscription ratio.

**Related concepts**
Configuration of Subscription Parameters
Cell Subscription Capacity
QCI Subscription Quanta

## 9.4 Configuration of Load Balancing Action on Node Level

The load balancing sensitivity and the maximum rate of load balancing actions are configured with parameters LoadBalancingFunction.lbThreshold and LoadBalancingFunction.lbCeiling. These attributes are defined at node level and apply to all load balancing relations within the node and towards target cells in other nodes. In many deployments, the default values of these attributes can be used.

Increasing the value of `lbThreshold` can reduce the rate of load balancing actions back and forth between cells as a result of random variations of the traffic load up and down. However, a certain degree of UE shuffling between the cells can be useful, as it reduces the risk of UE segregation due to differences in traffic behavior.

An increase of the value of `lbThreshold` can also increase the burstiness of load balancing actions. This means that load balancing occurs all at once rather than as a sequence of small steps.

Reducing the value of `lbCeiling` can help to distribute a sudden large amount of load balancing actions over a period of time. However, care is required to ensure that `lbCeiling` is configured large enough to outbalance any systematic drift in the load balance. For example, due to poor coverage-triggered mobility or due to mobility between neighbor cells on particular carrier frequencies.

Attribute `lbThreshold` can be configured greater than attribute `lbCeiling`. The effect can be that the load gap between the cells is not entirely closed.

## 9.5 Configuration of Measurements for UE Selection for Load Balancing Action

Inter-frequency Event A5 measurement reporting in the UE selection for load balancing action is configured for each source cell using the following attributes:

- `ReportConfigEUtraInterFreqLb.a5Threshold1Rsrp`
- `ReportConfigEUtraInterFreqLb.a5Threshold2Rsrp`
- `ReportConfigEUtraInterFreqLb.hysteresisA5`

For inter-frequency Event A5 measurement reporting, `ReportConfigEUtraInterFreqLb.lbA5Thr1RsrpFreqOffset` can be applied to `a5Threshold1Rsrp` to support a flexible `a5Threshold1Rsrp` between frequency relations.

To ensure that the UE can stay in the target cell after a load balancing action, configure `a5Threshold2Rsrp` attribute with a value higher than the Event A2 Poor Coverage threshold in the target cell. If QCI-dependent thresholds are applied in the target cell, parameter `a5Threshold2Rsrp` must be configured with a value higher than the maximum of these thresholds. The documents Coverage-Triggered Inter-Frequency Session Continuity and Service Triggered Mobility describe the configuration of poor coverage thresholds.

Attribute ReportConfigEUtraInterFreqLb.a5Threshold1Rsrp can be used to exclude UEs in a good radio environment from load balancing.

Attribute ReportConfigEUtraInterFreqLb.a5Threshold1Rsrp can also be used in a priority carrier configuration.

Attribute ReportConfigEUtraInterFreqLb.a5Threshold2Rsrq is configured for the source cell. It is used to set the RSRQ threshold for the target cell. The reporting quantity for Inter-Frequency Load Balancing is set to RSRP, so the RSRQ check is made at reception of an Event A5 measurement report. The load balancing action is ended if the reported RSRQ falls lower than the threshold.

Note: In a source cell where the Inter-Frequency Load Balancing feature is active, the UeMeasControl.sMeasure attribute must be set to disable the s-Measure threshold in the cell. Otherwise, the UE cannot perform the measurements required for the UE selection process.

QCI-dependent UE selection criteria can be configured for the individual E-UTRAN frequencies if the Service Specific Load Management feature is activated. For more information, see Service Specific Load Management.

## 9.6 Configuration of Rate Offset for Load Balancing

Attributes LoadBalancingFunction.lbRateOffsetCoefficient and LoadBalancingFunction.lbRateOffsetLoadThreshold are used to configure the rate offset mechanism for load balancing.

Attribute lbRateOffsetCoefficient controls the impact of the rate offset mechanism. The higher the value, the more biased load balancing is towards high capacity cells at low load. To switch off the rate offset mechanism, attribute lbRateOffsetCoefficient must be set to value 0. Otherwise, it is recommended to use the default value.

Attribute lbRateOffsetLoadThreshold controls up to which subscription ratio the rate offset is to have effect. The effect gradually decreases as the average subscription ratio increases and disappears at an average subscription ratio of lbRateOffsetLoadThreshold. The default value corresponds to an average subscription ratio of 0.5 and it is not recommended to go below this value.

**Related concepts**
Rate Offset

## 9.7 Idle Mode Mobility Aspects for Inter-Frequency Load Balancing Configuration

When the Inter-Frequency Load Balancing feature is used, consideration of the UE mobility in idle mode is recommended. The load balancing can be improved by appropriate configuration of the UE idle mode mobility. The appropriate configuration depends on the type of network deployment and cell topology.

Coverage-Triggered Inter-Frequency Session Continuity and Idle Mode Support provide additional information regarding the configuration of idle mode mobility.

### 9.7.1 Configuration of threshServingLow and threshXLow in Sticky Carrier or Priority Carrier Configuration

The recommendations to configure thresholds threshServingLow and threshXLow are the same for a cell deployment where the sticky carrier configuration is used and where priority carrier configuration is applied.

#### threshServingLow

Attribute threshServingLow (EUtranCellFDD.threshServingLow or EUtranCellTDD.threshServingLow) is used to set a threshold for the inter-frequency cell reselection at poor coverage. The recommendation is to configure threshServingLow attribute at the same level as the Event A2 Poor Coverage threshold in the cell. Event A2 Poor Coverage threshold configuration is described in document Coverage-Triggered Inter-Frequency Session ContinuityIdle Mode Support describes the configuration and function of idle mode mobility..

The sNonIntraSearch attribute in struct SIB3 for the E-UTRAN cell needs to be set to a value greater than or equal to threshServingLow attribute. In this case, the idle mode inter-frequency mobility operates sufficiently towards carrier frequencies with low CRP.

Note: The idle mode mobility thresholds, like threshServingLow, are defined relative to the required minimum RSRP of the respective cell.

The Event A2 Poor Coverage threshold is defined in absolute RSRP.

#### threshXLow

For a UE to reselect a cell on a carrier frequency with lower CRP, the RSRP of the target cell must exceed a threshold the EUtranFreqRelation.threshXLow attribute defines for that frequency. The recommendation is to configure threshXLow threshold with a small margin above threshServingLow threshold used in the cells on the target frequency.

**Note:** The `threshServingLow` parameter can be configured differently on different carrier frequencies. This must be taken into account when configuring the corresponding `threshXLow` attributes.

**Related concepts**
Sticky Carrier
Priority Carrier

## 9.7.2 Configuration of threshXHigh in Priority Carrier Configuration

For UEs to reselect a cell on a carrier frequency with higher CRP, the RSRP of the target cell must exceed a threshold that attribute `EUtranFreqRelation.threshXHigh` defines for that frequency. The recommendation is to configure threshold `threshXHigh` with at least a small margin higher than parameter `threshServingLow` used in the cells on the target frequency.

The value of threshold `threshXHigh` can be further increased to reduce the probability for UEs to select lower CRP carrier frequencies. This can be useful if the priority carrier cells are large enough to be selected by significant number of UEs. In this way, the bias achieved in the idle mode UE distribution can be moderated to a suitable level. The Inter-Frequency Load Balancing feature can then be applied to tune the UE distribution on top of that.

The possibility to exclude UEs in a good radio environment from load balancing can also be beneficial in a priority carrier configuration like the one shown in Figure 9.

The F2 frequency has higher CRP than the F1 frequency. This means that all UEs within the dotted lower circle perform idle mode cell reselection to the higher cell on frequency F2. There is a high risk that such UEs are load balanced back to the lower cell if the higher cell is more loaded than the lower cell. Avoid this by setting `ReportConfigEUtraInterFreqLb.a5Threshold1Rsrp` parameter so that only UEs outside the coverage area are subject to load balancing. Coverage are is defined by `threshXHigh` attribute.

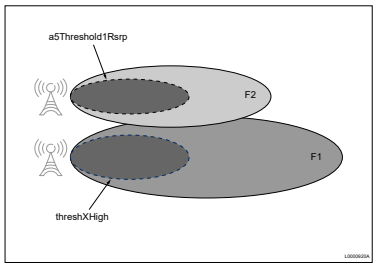



Figure 9 Example of Managing UE Load Balancing in a Priority Carrier Configuration

**Related concepts**
Priority Carrier

## 9.7.3 Combination of Idle Mode Mobility

The sticky carrier and priority carrier configurations can be combined in a multi-carrier network (one with more than two carrier frequencies).

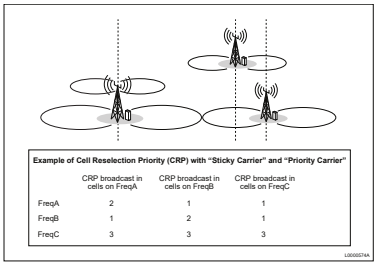



Figure 10 Combination of Sticky Carrier and Priority Carrier Configurations

The sticky carrier configuration is typically applied between carrier frequencies with similar coverage properties. The priority carrier configuration is suitable for carrier frequencies with substantial loss of UEs. The loss can occur because areas have poor coverage or have large coverage gap, and in areas where cells that are small compared to the cells on other carrier frequencies.

The way to configure the idle mode mobility can alternate between different locations within the network, depending on the local conditions.

Other principles for the configuration of the idle mode mobility are also possible. The Inter-Frequency Load Balancing feature is not restricted to particular configurations of UE idle mode cell reselection.

**Related concepts**
Sticky Carrier
Priority Carrier

## 9.8 Uplink-Heavy UE Detection Configuration

> Uplink-heavy UE detection can be enabled independently for the feature to improve the UE selection process for load-balancing mobility actions.

The following attributes must be set to `true` to enable uplink-heavy UE detection for load-balancing mobility actions:

- `EUtranCellFDD.lbUlHeavyUeDetectEnabled`
- `EUtranCellTDD.lbUlHeavyUeDetectEnabled`

The following can be used to configure uplink-heavy UE detection:

- `ENodeBFunction.ulHeavyUeDetectTime`
- `ENodeBFunction.ulHeavyUeBsrThresh`
- `ENodeBFunction.ulHeavyUeBsrHyst`

The attributes that configure uplink-heavy UE detection are the same attributes used by the Uplink-Traffic-Triggered Mobility and the Data-Aware Carrier Aggregation features. Therefore, changing these attributes also affects their uplink-heavy detection function.

# Appendix A: Feature Change History

> This section lists changes that affected this feature and the impact it had on the network.

## Appendix A.a: 19.Q1: Preferential Traffic Management

> The Preferential Traffic Management solution enables the operator to control user access to priority and non-priority bands. With Preferential Traffic Management, it is possible to define up to 12 subscriber groups denoted by a list of SPIDs and PLMN with different selection priority for Inter-Frequency Offload and Admission-Triggered Offload.

| | |
|---|---|
| Feature Name: | System Improvements in LTE RAN L19.Q1 |
| Feature Identity: | 20/FAJ 121 1447 |
| Value Package Name: | LTE Base Package |
| Value Package Identity: | FAJ 801 0400 |
| Node Type: | Baseband Radio Node, DU Radio Node |
| Access Type: | LTE |

The Preferential Traffic Management solution offers the operator the possibility to efficiently use an LTE network, deployed on several frequency bands, with one frequency band used primarily for priority users. The solution enables the following:

- To use the priority band for both priority and non-priority users as long as co-existence rules allow it.
- To use the priority band exclusively for priority users in the intended traffic situations.
- To use the non-priority bands for both non-priority and priority users, both as coverage bands and bands to offload priority traffic when the priority band is overloaded.
- To define how Preferential Traffic Management functions treat specific UEs by using subscriber groups.

The load management policies used depend on the current traffic load level. Traffic load states in the solution are defined by configurable load threshold parameters.

The solution is developed for existing commercial RAN infrastructure with a designated band used as priority band, using an MOCN architecture. The solution is achieved by a collection of extensions in existing licensed features, configured to work together.

For more detailed information on Preferential Traffic Management, see Preferential Traffic Management Guidelines.

### Capacity and Performance

The capacity on priority bands is limited for non-priority users and changes with network congestion level. This is because of the need to provide uninterrupted connectivity for priority users. During network congestion, an increased drop in mobility success rate towards priority cells is expected.

### Operation

The Preferential Traffic Management solution is activated on node level using the `PtmFunction.ptmEnabled` attribute.

On the eNodeB, a maximum of four cell profiles can be configured, but a cell refers to only one predefined profile. This controls which Preferential Traffic Management actions are triggered on the cell. Each cell profile can be configured using the `PtmCellProfile.cellType` attribute with the following settings:

- `PRIORITY`
- `NON_PRIORITY`

UE selection for Admission-Triggered Offload and Inter-Frequency Offload works only if the `PtmCellProfile.cellType` attribute is set to `PRIORITY`.

The function introduces the following new MO classes:

- `PtmAtoConfig`
- `PtmCellProfile`
- `PtmFunction`
- `PtmIflbConfig`
- `PtmIfoConfig`

- PtmResOpUseConfig
- PtmStmConfig
- PtmSubscriberGroup

This function is controlled by the following new parameters:

- PtmAtoConfig.minSuccRateThreshold
- PtmAtoConfig.ptmSubscriberGroupRef
- PtmCellProfile.cellType
- PtmFunction.ptmEnabled
- PtmIflbConfig.stopIncomingIflbThreshold
- PtmIflbConfig.stopOutgoingIflbEnabled
- PtmIfoConfig.ptmSubscriberGroupRef
- PtmResOpUseConfig.ptmSubscriberGroupRef
- PtmResOpUseConfig.resMsrUsageThreshold
- PtmResOpUseConfig.resSRatioThreshold
- PtmResOpUseConfig.unresMsrUsageThreshold
- PtmResOpUseConfig.unresSRatioThreshold
- PtmStmConfig.inhibitImpInterval
- PtmStmConfig.ptmSubscriberGroupRef
- PtmSubscriberGroup.plmn
- PtmSubscriberGroup.spidList

Handover of UEs with certain PLMN IDs and SPID to a target cell can be blocked. This occurs when the PLMN ID of the UE is "reserved for operator use" in the target cell. It can be reserved, because the priority cell has a very high traffic load—subscription ratio or MSR usage. It is also required that the SPID of the UE is configured in the PtmSubscriberGroup MO as the PtmStmConfig MO refers to that group.

The following features are affected by the solution:

Table 14 Affected Features

| Feature | Description |
|---|---|
| Admission-Triggered Offload, FAJ 121 3100 | Admission-Triggered Offload can be enabled per cell in a node to given target frequencies for UEs configured with a specific SPID. |
| Best Neighbor Relations for Intra-LTE Load Management, FAJ 121 3028 | New value IFO_AND_IFLBEUtranFreqRelation.lbBnrPolicy to allow automatic configuration for offload and load balancing actions. |
| Inter-Frequency Load Balancing, FAJ 121 3009 | Inter-frequency load balancing can be stopped for some cells—to priority cells with high traffic load from non-priority cells. This allows the priority band to be used primarily to serve priority users, and ensures access for priority UEs in high traffic load situations. is introduced for attribute |
| Inter-Frequency Offload, FAJ 121 3061 | Inter-Frequency Offload can be enabled per cell in a node to given target frequencies for UEs configured with a specific SPID. |
| Subscriber Triggered Mobility, FAJ 121 1788, RFPM function | Idle mode prioritization can be inhibited between some cells—to priority cells with very high traffic load from non-priority cells. This ensures that priority cells are not overloaded, so priority UEs have access in very high load traffic situations.<br>The offloadAllowed struct member in FreqPrioEUTRA struct is introduced to allow or disallow inter-frequency offload for UEs with a specific SPID to a target frequency. |

The following attributes are affected by the solution:

Table 15 Affected Parameters

| Parameter | Description |
|---|---|
| EUtranFreqRelation.lbBnrPolicy | New value IFO_AND_IFLB is is introduced for attribute EUtranFreqRelation.lbBnrPolicy to allow automatic configuration for offload and load balancing actions.<br>**Note:** The IFO_AND_IFLB value can only be used if Preferential Traffic Management is enabled and configured. |

| Parameter | Description |
|---|---|
| EUtranCellRelation.loadBalancing | New value IFO_AND_IFLB is introduced for the attribute to allow Inter-Frequency Offload and Inter-Frequency Load Balancing for the cell.<br>**Note:** The IFO_AND_IFLB value can only be used if Preferential Traffic Management is enabled and configured. |
| loadBalancingAllowed | A struct member in FreqPrioEUTRA struct of attribute RATFreqPrio.freqPrioListEUTRA. Offloading behavior can be controlled by offloadAllowed struct member in FreqPrioEUTRA struct, if loadBalancingAllowed is set to FALSE. |

### Interfaces

This change affects the following interfaces:

- X2
  New private information element for traffic load state of a cell is added to cell resource status report messages.

### Hardware

No special hardware requirements.

### Other Network Elements

To fully benefit from the change, the node requires support from the following network elements:

- ENM
  The enhancement requires ENM version 18.16 or later to be used.

## Appendix A.b: 19.Q2: Support for Load Balancing during Access Class Barring

> This feature enhancement enables Inter-Frequency Load Balancing to continue outgoing load balancing from overloaded cells with triggered Load-Based Access Barring.

| | |
|---|---|
| Feature Name: | Load-Based Access Barring |
| Feature Identity: | FAJ 121 3093 |
| Value Package Name: | High Load Handling |
| Value Package Identity: | FAJ 801 0415 |
| Node Type: | Baseband Radio Node, DU Radio Node |
| Access Type: | LTE |

The enhancement enables outgoing load balancing by calculating subscription ratio during triggered Load-Based Access Barring on the cell. This allows the overloaded cell to reduce the load faster and recover to normal state.

The function is supported on any cell, not only on priority cells defined by Preferential Traffic Management.

### Capacity and Performance

The main processor load is affected because of increased load balancing actions, signaling, and handovers.
The enhancement can trigger the RRC connection reconfiguration procedure.

### Operation

The enhancement introduces the following two new MO attributes:

- EUtranCellFDD.iflbAbEnabled
- EUtranCellTDD.iflbAbEnabled

### Interfaces

No impact.

### Hardware

No special hardware requirements.

### Other Network Elements

To fully benefit from the change, the node requires support from ENM version 19.04 or later.

## Appendix A.c: 22.Q4: Reserved Cell for Operator Use Improvement

> This system improvement enables priority UEs to select and access priority cells in high traffic load when the priority core network is not available.

| | |
|---|---|
| Feature Name | System Improvements in L22.Q4 |
| Feature Identity | 35/FAJ 121 1447 |
| Value Package Name | LTE Base Package |
| Value Package Identity | FAJ 801 0400 |
| Node Type | Baseband Radio Node |
| Access Type | LTE |
| Impact at Upgrade | Optional |

### Summary and Benefits

This system improvement enhances accessibility for priority UEs as part of the Preferential Traffic Management solution. It enables priority UEs that utilize non-priority equivalent home PLMN to connect to priority bands and to fully utilize priority cell resources under the following circumstances:

- The priority cells are in a high traffic load.
- The priority PLMN core network is not available due to S1-C connection failure.

### Capacity and Performance

This system improvement can be observed with the following PM counters:

- `EUtranCellFDD.pmPtmCellReservedMsr`
- `EUtranCellTDD.pmPtmCellReservedMsr`
- `EUtranCellFDD.pmPtmCellReservedSubRatio`
- `EUtranCellTDD.pmPtmCellReservedSubRatio`

### Operation

This system improvement introduces the `PtmResOpUseConfig.suspResOpUsePlmnList` attribute. This attribute can suspend dynamic PLMN reservation during S1-C connection failure. The operator can select priority PLMNs for which the cell reservation is suspended.

If the `PtmResOpUseConfig.suspResOpUsePlmnList` attribute is empty, which is the default value, any of the priority PLMNs can suspend dynamic PLMN reservation for cells during S1-C connection failure.

This system improvement introduces the following PM events:

Table 16 Introduced PM Events

| PM Event | Description |
|---|---|
| `INTERNAL_EVENT_PTM_CELL_RESERVED_START` | Triggered when the load is above threshold or the S1-C connection to the priority PLMN is recovered. It indicates that the cell is reserved for operator use. |
| `INTERNAL_EVENT_PTM_CELL_RESERVED_STOP` | Triggered when the load is below threshold or the S1-C connection to the priority PLMN failed. It indicates that the cell is no longer reserved for operator use. |
| `INTERNAL_EVENT_PTM_CELL_RESERVED_SUSP_START` | Triggered when the load is above threshold or the S1-C connection to the priority PLMN failed. It indicates that the cell reservation for operator use is suspended. |
| `INTERNAL_EVENT_PTM_CELL_RESERVED_SUSP_STOP` | Triggered when the load is below threshold or the S1-C connection to the priority PLMN is recovered. It indicates that the suspension of cell reservation for operator use is over. |

### Interfaces

No impact.

### Hardware

No specific hardware requirements.

### Other Network Elements

It is recommended to use ENM version 22.4 or later to support the MOM changes introduced by this system improvement. If the Release Independence Manager application is used, the MOM changes are supported on earlier ENM versions also.

**Related Information**
Preferential Traffic Management Guidelines
Admission-Triggered Offload
Best Neighbor Relations for Intra-LTE Load Management
Inter-Frequency Load Balancing
Inter-Frequency Offload
Subscriber Triggered Mobility
Shared LTE RAN

## Appendix A.d: 24.Q2.0: Uplink-Heavy UE Exclusion for Inter-Frequency Load Balancing

> This feature enhancement enables the exclusion of UEs marked as uplink-heavy from load-balancing actions.

| | |
|---|---|
| Feature Name | Inter-Frequency Load Balancing |
| Feature Identity | FAJ 121 3009 |
| Value Package Name | Multi-carrier Load Management |
| Value Package Identity | FAJ 801 0427 |
| Node Type | Baseband Radio Node |
| Access Type | LTE |
| Impact at Upgrade | Mandatory |

### Summary and Benefits

This feature enhancement introduces a mechanism to detect uplink-heavy UEs. If a UE is identified as uplink-heavy, it can be excluded from load-balancing mobility actions in FDD and TDD cells.
This feature enhancement has the following benefits:

- It is possible to detect uplink-heavy UEs for the Inter-Frequency Load Balancing feature.
- The UE selection process is improved for load-balancing actions.
- Ping-pong effect between cells for uplink-heavy UEs is reduced when the Inter-Frequency Load Balancing and the Uplink-Traffic-Triggered Mobility features are active at the same time in the network.

### Capacity and Performance

The feature enhancement decreases the number of uplink-heavy UE handovers triggered by the Uplink-Traffic-Triggered Mobility feature. This can be observed with the following PM counters in the source cells where the Uplink-Traffic-Triggered Mobility feature is active:

- `EUtranCellFDD.pmCellHoPrepAttLteInterFUlTraffic`
- `EUtranCellTDD.pmCellHoPrepAttLteInterFUlTraffic`
- `EUtranCellFDD.pmCellHoPrepSuccLteInterFUlTraffic`
- `EUtranCellTDD.pmCellHoPrepSuccLteInterFUlTraffic`
- `EUtranCellFDD.pmCellHoExeAttLteInterFUlTraffic`
- `EUtranCellTDD.pmCellHoExeAttLteInterFUlTraffic`
- `EUtranCellFDD.pmCellHoExeSuccLteInterFUlTraffic`
- `EUtranCellTDD.pmCellHoExeSuccLteInterFUlTraffic`
- `EUtranCellRelation.pmHoPrepAttLteInterFUlTraffic`
- `EUtranCellRelation.pmHoPrepSuccLteInterFUlTraffic`
- `EUtranCellRelation.pmHoExeAttLteInterFUlTraffic`
- `EUtranCellRelation.pmHoExeSuccLteInterFUlTraffic`

### Operation

The feature enhancement introduces attributes to control the detection of uplink-heavy UEs. When the following attributes are set to `true`, UEs marked as uplink-heavy are excluded from load-balancing mobility actions:

- `EUtranCellFDD.lbUlHeavyUeDetectEnabled`
- `EUtranCellTDD.lbUlHeavyUeDetectEnabled`

The following existing attributes are updated to control the detection of uplink-heavy UEs when the Inter-Frequency Load Balancing feature is not active:

- `EUtranCellFDD.ulHeavyUeDetectEnabled`
- `EUtranCellTDD.ulHeavyUeDetectEnabled`

Note: The `EUtranCellFDD.ulHeavyUeDetectEnabled` and the `EUtranCellTDD.ulHeavyUeDetectEnabled` attributes are only applicable when at least one of the following features is active:

- Uplink-Traffic-Triggered Mobility
- Data-Aware Carrier Aggregation Configuration

The following legacy attributes are used for specifying the configurations of uplink-heavy UE detection for load balancing:

- `ENodeBFunction.ulHeavyUeDetectTime`
- `ENodeBFunction.ulHeavyUeBsrThresh`
- `ENodeBFunction.ulHeavyUeBsrHyst`

When uplink-heavy UE detection for load balancing is enabled in the cell, the mechanism is applied to UEs that connect to the LTE cell through the following processes:

- Incoming handover
- RRC initial setup
- RRC connection re-establishment

### Interfaces

No impact.

### Hardware

No specific hardware requirements.

### Other Network Elements

It is recommended to use ENM version 24.2 or later to support the MOM changes introduced by this feature enhancement. If the Release Independence Manager application is used, the MOM changes are supported on earlier ENM versions also.

**Related Information**
Data-Aware Carrier Aggregation Configuration
Inter-Frequency Load Balancing
Uplink-Traffic-Triggered Mobility

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